

Prayas JEE 2025

Maths

DPP: 2

Trigonometric Functions

Q1 $\sin\left(\frac{\pi}{2} + \theta\right) \cdot \cos\left(\frac{3\pi}{2} + \theta\right) \cdot \tan\left(\frac{5\pi}{2} + \theta\right) \cdot \cot\left(\frac{7\pi}{2} + \theta\right)$

is equal to:

- (A) $-\sin^2 \theta$
 (B) $-\cos^2 \theta$
 (C) $\sin \theta \cdot \cos \theta$
 (D) $-\sin \theta \cdot \cos \theta$

Q2 Which of the following is positive number?

- (A) $\sin 1290^\circ$
 (B) $\cos 570^\circ$
 (C) $\cot 1200^\circ$
 (D) $\tan 960^\circ$

Q3 Find $\sin \theta + \sin(\pi + \theta) + \sin(2\pi + \theta) + \sin + (3\pi + \theta)$

..... upto 2012 terms.

- (A) -1 (B) 0
 (C) 1 (D) 2

Q4 Find

$$\cos\left(\frac{\pi}{2} + \theta\right) + \cos\left(\frac{3\pi}{2} + \theta\right) + \cos\left(\frac{5\pi}{2} + \theta\right) \dots \text{upto}$$

$$+ \cos\left(\frac{7\pi}{2} + \theta\right) + \dots$$

2021 terms.

- (A) $\sin \theta$
 (B) $\cos \theta$
 (C) $-\sin \theta$
 (D) $-\cos \theta$

Q5 Find value of $\sec(1140^\circ) + \operatorname{cosec}(1140^\circ) + \sin(1140^\circ) + \tan(1140^\circ) + \cot(1140^\circ) + \cos(1140^\circ)$

- (A) $\frac{5}{2}(\sqrt{3} - 1)$

(B) $\frac{5}{2}(\sqrt{3} + 1)$

(C) $\frac{5}{2}(\sqrt{2} + 1)$

(D) $\frac{5}{2}(\sqrt{2} - 1)$

Q6 Value of $\sin \frac{8\pi}{3} \cos \frac{23\pi}{6} + \cos \frac{13\pi}{3} \sin \frac{35\pi}{6} =$

(A) $\frac{\sqrt{3}}{2}$

(B) $\frac{1}{2}$

(C) 2

(D) 1

Q7 Value of $\cos 510^\circ \cos 330^\circ + \sin 390^\circ \cos 120^\circ$ is

(A) 0

(B) 1

(C) -1

(D) 2

Q8 $\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8}$ equals to

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{3}{2}$

(D) $\frac{3}{4}$

Q9 If $\operatorname{cosec} \theta - \sin \theta = m$ and $\sec \theta - \cos \theta = n$ then $(m^2 n)^{2/3} + (n^2 m)^{2/3}$ equals to

(A) 0

(B) 1

(C) -1

(D) 2

Q10 $\sin^2 17.5^\circ + \sin^2 72.5^\circ$ is equal to

(A) $\cos^2 90^\circ$

(B) $\tan^2 45^\circ$

(C) $\cos^2 30^\circ$

(D) $\sin^2 45^\circ$

Q11



The value of expression $\cos 1^\circ \cdot \cos 2^\circ \dots \cos 179^\circ$ equals

- (A) 0
(B) 1
(C) $1/\sqrt{2}$
(D) -1

Q12 The value of $\cos^2 \frac{\pi}{16} + \cos^2 \frac{3\pi}{16} + \cos^2 \frac{5\pi}{16} + \cos^2 \frac{7\pi}{16}$ is

(A) 2 (B) 1
(C) 0 (D) None of these

Q13 The value of $\cos(270^\circ + \theta) \cdot \cos(90^\circ - \theta) - \sin(270^\circ - \theta) \cos \theta$ is

(A) 0
(B) -1
(C) $1/2$
(D) 1

Q14 If $\frac{\sin 3}{\sin 5} = k$, then ' k ' is

(A) $k > 0$ (B) $k < 0$
(C) $k = 1$ (D) none of these

Q15 Simplify the following:

$$\frac{\sin(180^\circ + \theta) \cos(360^\circ - \theta) \tan(270^\circ - \theta)}{\sec^2(90^\circ + \theta) \tan(-\theta) \sin(270^\circ + \theta)}$$

Q16 Simplify each of the following:

$$\frac{\cos(2\pi + \theta) \operatorname{cosec}(2\pi - \theta) \tan(\pi/2 + \theta)}{\sec(\pi/2 + \theta) \cos \theta \cot(\pi + \theta)}$$

Q17 Simplify each of the following:

$$\frac{\operatorname{cosec}(90^\circ + \theta) + \cot(450^\circ + \theta)}{\operatorname{cosec}(450^\circ - \theta) - \tan(180^\circ + \theta)} + \frac{\tan(180^\circ + \theta) + \sec(180^\circ - \theta)}{\tan(360^\circ + \theta) - \sec(-\theta)}$$


Answer Key

Q1 (C)
Q2 (D)
Q3 (B)
Q4 (C)
Q5 (B)
Q6 (B)
Q7 (C)
Q8 (C)
Q9 (B)

Q10 (B)
Q11 (A)
Q12 (A)
Q13 (D)
Q14 (B)
Q15 $-\sin(\theta)(\cos(\theta))^2$
Q16 -1
Q17 2



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